Does a meta-doctrine system change model behaviour, or just make it legible? A self-ablation study under equal compute

Scott Greenwood

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Abstract

"Agentic" coding setups increasingly ship a *meta-doctrine layer*: always-on guidance—roles, gates, principles—together with a project record that grows across sessions, on the premise that this scaffolding improves the work. That premise is hard to test because of one confound: loading the guidance also loads *tokens*, and extra context alone can make a model more thorough. We therefore test one such system (the "Soul System") by re-running every apparent gain with the guidance **replaced by an equal volume of unrelated filler text**. A gain that survives the swap is real; a gain that disappears was the tokens, not the doctrine. Across ten doctrine probes and six instrument probes (Haiku 4.5, Sonnet 4.6, Opus 4.8; n = 5–7 per cell), this equal-compute control exposes four false positives, and what remains comes down to one principle: the system changes the model's *decisions* only when it supplies **project-specific knowledge or against-the-grain conventions that the model cannot reconstruct on its own and that a generic substitute gets wrong**. Recalling a recorded project fact avoided a trap 10/10 times, versus 2/5 with an equal-length irrelevant record; a recorded convention flipped an against-the-grain decision 10/10, where generic best-practice text gave the default answer 0/10. Everything aimed instead at *general reasoning lift*—forced roles (as prose *and* as dedicated sub-agents), trajectory-reading, compliance-resistance, and the handoff and verification *formats*—is reproduced by the filler. A synthetic multi-session test reaches the system's strongest claim: an against-the-grain fact recorded early and then buried under later work still prevented a dangerous drift in a much later session (5/5 → 0/5), and—alone among our results—this held even for the strongest model. Lacking the recorded fact, that model failed *more* confidently, inventing a non-existent API feature to justify the unsafe path. The organizing principle: **what the model can re-derive dissolves; what it cannot regenerate persists** (*derivable dissolves, unguessable persists*). **In short: the layer's measured value is memory, not reasoning.** It changes what the model decides only where it carries facts or conventions the model cannot regenerate; everything aimed at general reasoning is reproduced by equal-length filler and already supplied at the frontier. We give the resulting keep-or-cut decision and a ledger of what a single-interaction method cannot settle.

1 Introduction

A growing class of tooling wraps a language model in a *meta-doctrine layer*: standing instructions loaded into every session (personas or "roles," verification gates, design principles) together with a durable project record (decision logs, findings, conventions) that grows over time. The implicit and often-stated promise is that this layer *improves the work*—the model reasons more carefully, catches more defects, and stays coherent across a long project.

This promise is unusually hard to falsify by inspection, because the layer reliably changes the *surface* of the output: longer, more structured, more self-aware-sounding text. That surface change is consistent both with the layer genuinely improving the *substance* of decisions and with it doing nothing but making the model verbose and the work legible. Distinguishing the two requires a control the field rarely applies.

We study one mature instance—an open meta-doctrine system we call the Soul System, comprising $\sim$16 commands, a three-layer always-on doctrine, and an append-only record—and ask a deliberately skeptical question: **which parts change what the model decides, and which only change how the work reads?** Our contributions are:

1. **An equal-compute ablation protocol** (Section 3) that re-runs every apparent win against a length-matched, semantically-empty filler, isolating doctrine-substance from the compute that any long prompt supplies.
2. **A behavioural map** (Section 4) across doctrine and instruments, capability tiers, and task structures, including four wins that the control falsifies.
3. **A single organizing axis**—*derivable-dissolves, unguessable-persists*—that predicts what survives, and a synthetic multi-session result showing the surviving value is the first in the study that does not dissolve at the frontier.
4. **A keep/cut decision** for a lean version (Section 5) and an explicit ledger of what a single-shot method cannot measure (Section 6).

The result is not a defence of the system and not a debunking. It is a separation: a narrow, real value that holds even for strong models—carrying the unguessable—from a larger body of legibility that a generic equivalent matches.

2 Related work

Context quantity as a confound, and what filler does. The premise of our control is that added tokens can change a model's output independent of their content. The direction of that effect is not simple, and the literature is split in a way that matters for interpretation (Section 6). Pfau et al. [14] show that even *semantically-empty* filler tokens can supply usable parallel computation—small transformers solve problems with filler they cannot solve answering immediately—though the ability is hard to acquire and was absent in commercial models of that era. In the opposite direction, and more robustly, *irrelevant* context **degrades** reasoning: Shi et al. [15] ("distracted by irrelevant context") find accuracy drops sharply when in-topic but irrelevant information is added. So extra tokens are not free, and their sign depends on model and content—which makes a length-matched filler the right instrument but its bias direction something to argue, not assume.

Roles and personas. Assigning static personas in system prompts does not reliably improve accuracy; Zheng et al. [1] test 162 roles across model families and find no general benefit, and Kim et al. [7] show role-play *degrades* reasoning on a third of datasets even at the frontier, with bidirectional, near-zero net effect. The boundary: Kong et al. [8] find an engineered multi-turn *role-immersion* prompt helps zero-shot reasoning—but as an implicit chain-of-thought trigger, a different intervention than a static persona. This matches and bounds our doctrine result (Sections 4.1, 4.3).

Single- vs. multi-agent decomposition. Under matched compute the case for multi-agent orchestration weakens. Tran and Kiela [2] show single-agent LLMs *outperform* multi-agent systems on multi-hop reasoning at equal thinking-token budgets and argue reported multi-agent advantages are "better explained by unaccounted computation"; Gao et al. [3] find the advantage diminishes as base models improve. Notably the canonical multi-agent-debate result [9] reports gains *without* a compute-matched baseline—the exact confound our method targets. Our agentic-roles probe (Section 4.3) reaches the same conclusion for Soul's roles rebuilt as sub-agents.

Knowledge vs. reasoning; retrieval over parametric memory. Our organizing axis—*derivable dissolves, unguessable persists*—restates a robust result. Lewis et al. [10] (RAG) show parametric models have "limited ability to access and precisely manipulate knowledge" and that adding a non-parametric retrieved store beats the parametric model; Ovadia et al. [11] find retrieval "consistently outperforms" unsupervised fine-tuning for injecting facts the model lacks. Gan and Sun [4] (RAG-MCP) extend the same retrieve-don't-preload logic to tool selection, underwriting both our never-load-everything doctrine and the skill-routing probe (Section 4.4).

Calibration and confident fabrication. Our longitudinal result turns on a model *fabricating* a missing fact rather than abstaining. The GPT-4 technical report [12] documents that the model "can be confidently wrong... not taking care to double-check when it is likely to make a mistake," and that RLHF post-training *degrades* calibration (expected calibration error rising roughly tenfold from the pre-trained model). Kadavath et al. [13] show models have partial but imperfect self-knowledge of what they know. We return in Section 6 to what this does and does not license about *scale*.

Positional use of long context. Liu et al. [5] ("lost in the middle") show models use information at the beginning and end of long inputs far better than the middle—the degradation a record-burial probe must control for, and a caveat on our depth result (Section 6).

Self-critique limits. Verification gates assume a model can catch its own errors; Huang et al. [16] find LLMs "cannot self-correct reasoning yet" without external feedback—consistent with our result that the gate *concept* helps but its elaborateness does not (Section 4.2).

Persistent memory and rules. External-memory agents (MemGPT [17]; generative agents with a memory stream [18]) make the cross-session-record hypothesis concrete, though neither isolates record-content from compute as we do. Constitutional AI [6] steers behaviour from a small explicit rule-set rather than dense supervision; the Soul System's compressed always-on rules are a project-level analogue (Section 4.2).

We position this as an *engineering self-ablation*, not a survey: its contribution is the equal-compute discipline applied uniformly and the longitudinal harness, not breadth of coverage.

3 Method

3.1 Arms and models

The unit of measurement is an **arm**: a tuple `(task, condition, model, run-index)`. Each arm is an isolated, non-interactive model call (`claude -p "<task>" < /dev/null`, fresh process, no chat history); the thing under test (doctrine, a role lens, a gate checklist, a handoff cursor, a record) is supplied via the system prompt or pasted into the task, and the bare arm receives the task alone. We run $n \geq 5$ per cell unless noted, across a capability range—Haiku 4.5 (weak), Sonnet 4.6 (mid), Opus 4.8 (frontier)—because how much any aid helps depends on whether the model already succeeds without it.

3.2 The three controls

C1 — Equal-compute control (decisive). The dominant confound in this class of claim is that loading the system also loads tokens. Every apparent win is re-run against a **length-matched, semantically-empty filler** of the same budget: ∼26 KB of unrelated warehouse-logistics prose in place of ∼26 KB of doctrine; a generic "double-check your work" in place of a named gate checklist;

a plain summary in place of a structured cursor. *A substance verdict that vanishes under equal compute is not substance.* C1 caught four false positives here (Section 4.1–4.2).

A caveat on direction (developed in Section 6). The filler is not neutral padding: the literature shows that added irrelevant tokens more often *degrade* a model than improve it [15], while empty filler can occasionally *supply* extra computation [14]—and both effects depend on the model and the content. That makes C1 a **conservative** control, not a symmetric one. A gain that *survives* the filler is real with room to spare, since the filler, if anything, worked against the comparison arm. But a gain that *vanishes* is ambiguous: it may have been the extra tokens, or the filler may have actively hurt the control arm. Our four filler types are either *coherent-but-irrelevant* (warehouse-logistics prose) or a *generic nudge* ("double-check"); where they tied the doctrine they usually also beat the bare baseline (Section 4.2), so they were not degrading in those cells. The clean separation—a three-arm control that adds a coherent-irrelevant distractor alongside empty padding—is run in Section 4.6, and it confirms on our models that coherent-irrelevant filler is not neutral, while empty padding behaves like bare ($\approx$ bare).

C2 — Difficulty validation. A task measures an aid only if the bare baseline plausibly fails it. Before scoring, the bare arm is run $n \geq 5$; if it already succeeds, the task measures ceiling, not the aid, and is discarded or sharpened. A corollary learned the hard way: a high-salience domain (e.g. patient safety) can trigger a caution reflex that *substitutes* for the discipline under test, passing the baseline for the wrong reason; one medical task was discarded for exactly this. Domains are kept low-salience.

C3 — Trap taxonomy. When a task plants a defect, the defect is of a named kind so a "catch" maps to a specific discipline: a false premise (anchoring), a dropped requirement (completeness), a buried downstream consequence (trajectory), a silently-truncated long-horizon constraint (gate). One trap per task.

3.3 Scoring

Rubrics and answer keys are **pre-registered** before outputs are seen. Outcomes are scored mechanically where possible (count the behaviour; within-cell comparable keyword patterns) and then **read-confirmed** on borderline arms. Counts are never eyeballed, and we explicitly watch for *wrong-reason* passes (an artifact refused for missing information, a placeholder flagged, a disaster-salience reflex)—these are designed out or scored separately. One pass in this study *looked* like a decisive win on a keyword count (5.8 vs. 2.6) and reversed on the binary read (both 5/5): the count had measured verbosity of flagging, not catches.

3.4 Three honest value buckets

Not everything is single-shot A/B-testable, and faking a single-shot proxy for a longitudinal or human-workflow claim would be the precise self-consistent-but-false result the study exists to catch. Each component is assigned a bucket: **Measured** (A/B + C1), **Reasoned** (friction/convenience, assessed by transparent cost-vs-value argument, not a faked experiment), or **Inspected** (one-time/structural artifacts).

3.5 Harness and a contamination finding

Arms run from a scratch working directory; condition files are supplied via the model CLI's system-prompt-file flag or pasted inline. **Cross-file `@`-imports are not used** under the non-interactive mode: they fail silently and the model confabulates the missing content ~43% of the time (this is itself one of the project's findings, and the deciding knowledge in the recall probe), so content is inlined and a sentinel load-test confirms it reached the model before any run is trusted. A separate contamination was caught and fixed: non-interactive arms have tool access and read the working directory, so arms launched from a directory containing the answer file simply *grepped it* even in the no-record condition; every arm is therefore run from an **isolated empty directory**, and the contaminated run is retained as evidence.

The harness, the per-experiment pre-registrations and results, and the scoring spec are packaged as a reusable benchmark—the *meta-doctrine ablation suite*—in its own repo `Soul-Benchmark` (extracted from this repo at the 1.0 release), so the experiments in Sections 4 and 6.1 can be re-run and others can plug in their own doctrine and models.

3.6 Known bounds of the method

The harness is single-shot / single-resume; the system's strongest claim is intrinsically longitudinal, reached here by a synthetic session-chain harness (Section 4.5) rather than real project history. n is small (5–7/cell); keyword scoring is within-cell comparable, not absolute. The scorer is also the experimenter; rubrics are pre-registered to constrain this, and where bias is plausible it runs *toward* the system, so null results are conservative.

4 Results

Results at a glance. Every row is run against the equal-compute filler control (C1); a result that vanishes under C1 was tokens, not doctrine. "Survives" = the gain remains under the length-matched control. (Full designs and data: Sections 4.1–4.6; consolidated tables in `data/`.)

Probe	Control	Key result	Survives?	Verdict
Doctrine: roles / gates (frontier)	C1	changes *form* (named, legible), not substance	— (form)	lean to a line
Doctrine: hard regime, weak baseline	C1	substance only on **anchoring** (filler 0/3)	**yes** (narrow)	KEEP anchoring
Record recall (unguessable fact, F038)	C1 + 3-arm	**record 10/10** vs filler ≈ 0; isolates cleanly	**yes**	**KEEP — clearest win**
Convention transmission (docs-near-code)	3-arm	flips an against-the-grain decision; but derivable + primable	**yes** (narrower)	KEEP content, moderated
Roles as sub-agents	C1	A2 ≈ C1; multi-agent ≤ single; synthesis drops findings	no	form, prose→agents
Skill routing	oracle vs self-pick	**oracle 5/5 vs budgeted 1/5** (weak)	**yes** (bounded)	KEEP tested catalog
Longitudinal carry (7 rungs)	C1, synthetic chain	drift **5/5→0/5**; **persists at the frontier** (fabricates when missing)	**yes**	**KEEP — strongest**
Verify gate (high + low stakes)	3-arm ×2	bare ceilings; doctrine adds only a citation (apparent stabiliser did not replicate — n=10 rerun)	no	form / legibility
Handoff cursor	3-arm (cursor vs prose)	**cursor 10/10 = prose 10/10**	no	form / legibility
Calibration (Haiku→Sonnet→Opus)	3-tier ladder	fail-to-recall flat (robust); confident-fabrication *gradient* did **not** reproduce (scoring-fragile)	n/a	non-elimination only (Fig. 1)
Depth + position	token-scale burial	**30/30**, no middle decay (unique needle)	n/a	bounds the depth claim

Derivable dissolves; unguessable persists. The detailed sections follow.

4.1 Doctrine (the always-on layer)

Against a careful frontier baseline, removing a single role or a single gate changed the *form* of the work (it became named, legible, auditable) far more than its substance or reliability. A deliberately hard set of tasks—a buried logic trap, a false premise pushed under authority pressure, and a ten-constraint long-horizon task—was also form, not substance, at the frontier (removing the completion gate on the long-horizon task changed nothing: gate $\approx$ bare).

At a **weak baseline** (Haiku), with the full doctrine loaded and C1 applied throughout, the doctrine adds real substance in exactly one place: **challenging an unverified premise** ("anchoring"). It lifted the false-premise task from 0/3 to ∼1.5/3 while the filler stayed at 0/3, and it replicated on a second anchoring task (full doctrine 3/3 vs. filler 0/3). This substance is *content-dependent* (generic thoroughness cannot fake it), appears *only at a weak baseline* (a mid-capability model already self-corrects), and *depends on task structure* (tasks that invite analysis self-correct earlier than tasks that invite compliance).

Everything else dissolved under C1, including three apparent generalizations: trajectory-reading (bare 1/3 → full doctrine 3/3, but filler 2/3—generic long context did the work), compliance-flagging (the filler reproduced it—a matter of disposition), and stakes-dependent withholding (filler 5/5 = full doctrine—disposition, not doctrine). Without the equal-compute control, the study would have claimed an edge in frontier-model compliance that does not exist. This lines up directly with Zheng et al. [1] (roles do not lift accuracy).

4.2 Instruments and the record

Handoff/resume (Measured; form). A fresh session was given a curated snapshot of the record plus one of four resume notes—a structured cursor, a length-matched generic prose summary, a terse compact note, or nothing. Any rich note surfaced more constraints than a cold start ($\approx +2$ constraints; 0 scope-invention errors vs. 1)—but the *Soul-specific format is just form*: a length-matched generic summary tied or beat the structured cursor (on Sonnet the generic note surfaced 4.8 constraints on average vs. the cursor's 4.6; on Haiku the two tied at 4.6, both out of 5). The richness of the note matters; its structure does not.

Record-recall (Measured; KEEP — the clearest win). A trap task whose deciding piece of knowledge is a real, non-obvious finding from the project was run with three conditions: the relevant record, no record, and a length-matched *irrelevant* record (C1). The relevant record produced trap-avoidance **10/10**; the irrelevant record **2/5**; no record 2–3/5 (capable arms partly reason toward the right caution but mostly still recommend the trap). The *specific recorded content*—not "having a record," and not the extra tokens—is what makes the model reliably correct.

Verification gate (Measured; concept yes, format no). We asked the model to finalize a draft that contained one planted defect—an unsupported claim. Bare caught it 2/5; *any* check (a vague generic one, a generic checklist, or the Soul checklist) lifted this to ∼5/5. The gate *concept* earns its place, but the Soul-specific checklist does **not** beat a generic "double-check carefully" on catch-rate (both 5/5)—it only adds itemized legibility, which is form. (A second defect, a dropped requirement, was a C2 ceiling: 5/5 everywhere.)

Distillation / the "Mind" (Measured; KEEP the content). Here the deciding item is a decision where the project's convention *goes against the model's trained-in default*: keep docs next to the code, versus the default of a separate docs tree. Loading the project's rules—either the compressed "Mind" or a raw excerpt of the records—flipped the recommendation to the project's answer ∼10/10; a length-matched set of *generic principles* gave the opposite, default answer ∼0/10. This is the same axis as recall: carrying an against-the-grain convention that a generic equivalent gets wrong. The compressed Mind transmitted the convention no better than an equal-size raw excerpt (`Mind` ≈ `raw`), so the rule's *content* is the substance; whether it is better produced by auto-distillation or by hand-curation is not settled here. **Moderation (Section 4.6):** under a stricter three-arm control, this particular convention turned out to be partly *derivable* and primable by disposition—the two-arm comparison overstated the gap because its control text was a loaded counter-example. The recall result (an unguessable *fact* that a generic equivalent gets confidently wrong) is the stronger, cleaner member of this KEEP; the convention is real but narrower.

4.3 Roles as agents

A natural objection to "roles = form" is that the roles were tested as prose, never as *dedicated sub-agents*. Rebuilt as agents—one sub-agent per Council role, plus a synthesis step—Soul's roles still only matched a generic decomposition at equal compute, and the multi-agent setup did not beat a single agent told to work through the same perspectives in one context (it tied at best, and lost where the synthesis step pruned valid findings). The substance is *breaking the problem into explicit perspectives*—which is cheap, generic, and works in a single context—not the Soul framing or the multi-agent orchestration. The "form" verdict therefore extends from prose to agents, consistent with Tran and Kiela [2] and Gao et al. [3].

4.4 Skill routing (a bounded win)

One capability close to reasoning survived the control: *knowing which specialized skill to reach for*, when the model can select only a few. Given a catalog that held a tempting-but-wrong trap skill alongside a non-obvious correct one, a weak model forced to pick ≤ 2 reliably grabbed the trap and missed the right skill (self-pick 1/5); handed the right skill outright, it solved the task 5/5 (given-the-answer vs. budgeted self-pick = 5/5 vs. 1/5 at Haiku). The effect dissolves at the frontier (the strong model already knows the answer) and without a budget (free selection over-picks and catches the right skill anyway). It thus lives in the same cell as anchoring—*weak baseline* × *against-the-grain knowledge* × *tight budget*—and is the deployment side of the record; the number is a best-case (perfect-routing) upper bound. The retrieval framing matches Gan and Sun [4].

4.5 Longitudinal accumulation (the strongest claim)

The system's central promise is inherently about many sessions and cannot be faked in a single interaction. We reach it with a synthetic **session chain**: an early session records an against-the-grain decision D (together with the incident that established it); later sessions bury D under unrelated entries; then a fresh session, with D's original context gone, faces a new task that tempts the dangerous default. *Drift* means the default returns. Outcomes are read from the code the model produces, not from keywords.

Carrying the fact prevents drift, and this holds even at the frontier. With D recorded and carried forward, drift fell **5/5 → 0/5**; a length-matched filler record drifted exactly as if there were no record at all (C1 passes). Crucially, this is the **first win in the study that does not dissolve at the frontier**: the strong model drifts *too*, and worse—lacking the recorded fact, it does not merely fall back to the default, it *invents* a reassuring feature that does not exist (a payment idempotency key the API does not have) and explains why that makes the unsafe retry safe. That is a confident fabrication rather than an abstention, consistent with the loss of calibration documented for post-trained frontier models [12]. The mechanism is that D is knowledge the model *cannot* *regenerate*, not a reasoning skill it can. (In Section 6 we are careful not to over-read this as a *monotonic* "stronger ⇒ more confident fabrication" scale law—our own data below shows the failure direction depends on the rule's *form*, not on capability alone.)

Facts and conventions both carry; the dividing line is derivability. A companion probe recorded an arbitrary *convention* (return a Result type, never raise) rather than a fact; it too persisted at the frontier—the strong model deferred to the recorded convention instead of overriding it with its own habit. The organizing line is therefore **derivable dissolves, unguessable persists**: what the model can re-reason its way to (generic anchoring, routing at the frontier) dissolves under the control; what it cannot regenerate (a fact *or* an arbitrary convention) persists.

The carry holds even when the fact is buried deep. Varying how deep the fact sat in the record, it still worked 5/5 when buried under 20 intervening decision records, at both capabilities, with no sharp drop-off through $N = 20$ (the frontier control still drifted at depth). This is exactly the regime where lost-in-the-middle degradation [5] would predict failure; within the depth we tested, it did not appear.

Compression preserves the rule but not the force of the fact behind it. Distilling D down exposes a limit that matters for any always-on rule-set. The *directive* itself survives compression to a single line (0 violations across 45 cells, at both capabilities). The unguessable *fact* behind it does not: strip the incident and the explicit "no," leave only the terse rule, and the frontier model fabricates a replacement justification—on a gradient set by how much of the original force survived (full record 0/5, faithful distillation 2/5, aggressive erosion 4/5), while the weak model never fabricates (0/15—it lacks the confident prior). A second, independent counter-prior fact reproduced the gradient but *reversed* the capability direction (the weak model drifted where the frontier held). A further probe—holding the prior fixed and varying only the *form* of the directive—pinned down the mechanism: the frontier model flips $0/5 \rightarrow 5/5$ when an imperative rule is replaced by one with a *loophole* ("reuse only if validity is confirmed"), fabricating the missing feature to walk through the opening. So **the form of the directive governs the frontier model, and the strength of the prior governs the weak model.** The practical consequence (adopted as a doctrine amendment): when compressing a fact that contradicts a strong model's prior, preserve its *force*—the incident, the explicit negation—not just its statement, and keep the surviving directive imperative and free of loopholes.

The carry has also paid off in real use, not only on the synthetic test. The chain above is synthetic *by necessity*: the counterfactual—what a later session does *without* the record—exists only if we manufacture the no-record branch, since real history only ever runs the with-record branch. What real history offers instead is observational evidence. Six instances in the project's own 143-entry record show the accumulated record or a gate catching the project's *own* drift in live work: a fabricated rule-citation in an experiment's measurement, a retire-audit whose anchor was scoped too narrowly to see a live reference, a citation over-claim, two over-stated reproducibility results, and one decision that a blind run of the same model got wrong but a record-carrying run got right (SOUL-144). These lack a controlled counterfactual and are a *census*, not a rate—there is no count of the times a gate could have fired and did not—but they confirm that the mechanism has already fired on real multi-session history, which the synthetic chain cannot reach. They measure the payoff *to the agent* (a gate or finding catching drift); the payoff to a *human* in legibility over time remains asserted, not measured (Section 6).

4.6 The three-arm control (separating compute, distraction, substance)

Motivated by the literature in Section 2, we ran the three-arm version of C1—adding **empty padding** (length-matched lorem ipsum, i.e. pure compute) and **coherent-but-irrelevant** prose (length-matched off-topic operations text, i.e. distraction) alongside the bare arm and the substance arm. We ran it on five test vehicles in all; the first two were chosen to differ on a single dimension—whether the project's answer is *derivable*—and that pair is the clearest single illustration of the study's organizing thesis.

On a **derivable convention** (the project keeps documentation next to the code, versus the common default of a separate `docs/` tree), the yes/no outcome did *not* isolate substance. The frontier model already gave the project's answer at baseline (it re-derives "central docs drift"), and at the weak baseline a *coherent-irrelevant* filler drove the project's answer 5/5—its disciplined, businesslike tone primed a cautious disposition that re-derived the same conclusion—while empty padding sat at the bare level (2/5). Only the *reasoning* set the substance arm apart (it cited the specific recorded incident; the distractor re-derived the answer generically). This directly confirms, on our own models, Section 2's warning that coherent-irrelevant tokens are not neutral, and it **moderates the convention-transmission result of Section 4.2**: a project answer that is also a defensible generic answer is partly derivable and can be primed by disposition, so its two-arm "win" overstated the substance actually isolated (the original generic-principles control there was a *loaded* counter-example, which inflated the gap).

On an **unguessable fact** (a recorded finding that a certain import mechanism fails silently in non-interactive mode, and the model makes things up around it—the recall probe of Section 4.2), the same three-arm control isolated substance **cleanly**: the record produced the correct, trap-avoiding recommendation **10/10**, while bare, empty padding, and coherent-irrelevant filler each recommended the trap **0/10**. Here the coherent filler was *harmless* ($\approx$ bare)—the opposite of the convention probe—because no disposition can re-derive a fact the model simply does not have. Two side confirmations: empty padding $\approx$ bare on both probes (pure length did not move these decisions on our models), and the frontier model, lacking the fact, *confidently asserted the opposite* ("imports resolve in non-interactive mode the same as interactive"—false), a second, separate instance of the confident fabrication from Section 4.5.

The pair is the thesis in miniature: a coherent-irrelevant filler reproduces a *derivable* answer and is powerless against an *unguessable* one. The recall (record) KEEP survives the stricter control; the convention KEEP is real but narrower than its two-arm number suggested.

A third probe extends the control to a **"form" verdict**—the completion/verify gate ("passing the local tests is not the same as global correctness; check the property the whole system must satisfy before declaring done"). On a high-stakes vehicle (a payment-splitting refactor with green unit tests, asked "ship it?"), the three-arm control did not merely moderate the gate—it found it *fully derivable*: **all four arms avoided the trap 40/40**, bare included. With no injected context, both models already withhold the ship and demand real-world validation (a reconciliation check, shadow-replay against production data, a canary release). The substance arm's only isolable contribution is a *citation*: it names the specific doctrine (6/10 cells) where no other arm does, and it lifts the explicit "invariant" framing from re-derived (bare 7/10) to universal—while leaving the decision itself unchanged. The verify gate is thus the *most* derivable verdict we measured, more so than the convention (whose bare baseline at least sat at 2/5): its value is **legibility** (a named property to check, an auditable citation), not a change in behaviour, because the disposition it records is one the model already holds. The three probes form a spectrum of the organizing thesis—from an *unguessable fact* (recall, clean isolation, 10/10 vs 0), through a *derivable convention* (docs: moderated, primable by disposition), to an *already-held disposition* (verify: fully derivable, at ceiling). One limitation of the verify probe—that its high-stakes vehicle maximally primes caution (financial correctness)—was then tested with a low-stakes vehicle (an internal analytics dashboard, where shipping on green tests is defensible). Bare still withheld 10/10: caution is re-derived even away from the high-caution domain, so "legibility, not behaviour" survives the stricter test. An *apparent* residual behavioural effect—a coherent-irrelevant distractor lowering Sonnet's caution to 3/5 at $n = 5$ (two cells flipping to "ship now, verify post-deploy")—did **not** survive replication: a rerun at $n = 10$ returned 10/10 (no erosion), with Haiku unchanged at 5/5, so the 3/5 was small-sample noise, not a real effect. Away from the high-caution domain, then, the low-stakes verify gate is *purely* legibility—no behavioural effect survives the control. (Coherent-irrelevant filler is still not neutral: on the *derivable* convention probe it *primed* the answer 5/5, and on the *unguessable* fact it was harmless—its sign depends on the task—but it did not erode vigilance here once re-run; cf. [15].)

The last "form" verdict, **handoff**, closes the same way and completes the picture. Testing the actual handoff claim—does a structured resume *cursor* beat an equal-length *prose* summary carrying the same state?—the two tied exactly (10/10 = 10/10): both resumed flawlessly (correct next action, every live constraint restated), while the no-state arms scored 0/30. The cursor's structure is legibility; the state *content*, which prose carries equally well, is the substance. A side-result sharpens the fabrication thesis: the no-state arms did not *invent* a resumption—all 30 recognised the empty context and asked for the state. Confident fabrication (Section 4.5) fires when the model has a plausible prior to make something up from, not when the gap is obvious; a visibly empty context makes the absence plain, so the model abstains. With both "form" verdicts now resolved as legibility, **no "form" verdict survives the control as a change in behaviour.**

5 The lean-down: what survives

Applying a default-to-cut rule (keep only measured value, or a clearly argued unique value that a generic equivalent lacks), with one counterweight—*scaffolding that makes a good practice reliably happen has value even when its content is not unique* (a completion hook that fires; a capture command that lowers the friction of recording; a format that makes knowledge findable)—the system reduces to a small core:

1. **The records** (append-only log, findings, ideas): the proven carrier of specific, otherwise-underivable knowledge. *Keep — the core.*

2. **A minimal always-on seed**: the framing principle, the anchor obligation (the one bit of reasoning substance), the project's against-the-grain conventions as a short rule-set (the distill result), and one-line expansion / ask-the-human nudges. No role-chamber prose.

3. **A completion gate**: a short pre-ship check whose load-bearing item is the anchor check, fired as a hook (activation over prose).

4. **One capture command** (plus an earned-finding gate) and setup.

5. **A tested, selective skills catalog** with budget-aware routing—the deployment side of the record, valuable where weak models face selection pressure.

Cut, or folded into a one-line convention: the forced-perspective council (form, both as prose *and* as agents), the role-as-subagent engine, the handoff-cursor *format*, the read-only utilities, and the elaborate verification checklist. The bet is explicit: the value is against-the-grain project knowledge in three forms—*having* it (recall), *compressing* it (the seed rule-set), and *deploying the right piece when* it is not obvious (routing)—plus the anchor discipline and the activation scaffolding, all concentrated where the generic default is wrong; the frontier model dissolves the reasoning-ceremony because it already knows. This is what the evidence supports retaining, and what it supports retiring.

6 Threats to validity and what was not settled

The method's deliberate blind spot is its honesty test. Several of the largest bets sit on the side it cannot measure in a single interaction, and faking a single-interaction proxy for one of them would be the exact self-consistent falsehood the study exists to catch.

- **Longitudinal accumulation** is reached by a *synthetic* session chain—which supplies the controlled counterfactual that real history cannot—now corroborated by six *observational* real-world catches on the project's own record (Section 4.5; SOUL-144), which supply real multi-session history but not the counterfactual. Seven aspects are tested (carry, convention, depth, erosion, weak distiller, a second prior, form-vs-prior); scale (many interacting facts, a long eroding record) remains a broadening of the harness still to be done.

- **Multi-task routing** (does a curated catalog pay off across a range of tasks?) and a **built router** (versus the best-case, given-the-answer bound) are unmeasured; the routing win rests on one task and a perfect-routing upper bound.

- **Tool-skills** (giving arms genuinely different real tools) cannot be granted to headless arms; the routing probe used procedure-knowledge as a faithful proxy for the core selection problem.

- **Retrieval in a large repo** could be closed, but was deferred as low marginal value (it is predicted to simply re-confirm that "the record carries the value").

- **Legibility's payoff to a human over time**—the standing justification for keeping the record readable—is asserted, not measured; it is inherently human and longitudinal.

- **The filler control is conservative, not symmetric.** As Section 3.2 notes and the literature establishes, irrelevant tokens more often degrade a model than inflate it [15], so a vanished gain confuses "it was just compute" with "the filler hurt the control." This biases the study toward *under*-stating the doctrine, which makes the surviving wins (recall, distill, longitudinal) robust but leaves some "form" verdicts with a residual ambiguity. The three-arm control (Section 6.1) was run to address this; it confirmed the recall KEEP, moderated the convention KEEP, and resolved both "form" verdicts—verify (derivable) and handoff (a structured cursor ties equal-length prose)—as legibility, not behaviour. No residual remains on this axis except the human-legibility-over-time payoff, which is longitudinal and unmeasured (below).

- **"Stronger $\Rightarrow$ more confident fabrication" is not claimed as a scale law — and the apparent gradient did not reproduce.** The established result [12,13] is that capability does not *eliminate* confident fabrication, and that RLHF degrades calibration—a training-incentive effect, not a smooth function of scale. The cross-scale calibration probe (Section 6.1) tested this on a three-tier ladder. *What survives a rerun:* recall of the unguessable fact does not improve with capability (no tier spontaneously recalls it—flat across tiers), and capability does not *eliminate* the confident false assertion (the top tier still states it in a real fraction of cells). *What does not survive:* the **gradient** (Haiku $<$ Sonnet $<$ Opus, strongest states it most). A rerun under stricter scoring moved the top tier from $\sim$80% to $\sim$0%, because the count is fragile to how it is scored: the strongest model tends to assert the falsehood *and then* recommend verifying it, and it rejected the shortcut outright 9/15 (SOUL-142). The honest claim is *non-elimination only*—not a scale law on either the yes/no measure (saturated) or the confidence rate (unstable).

- **The depth result, now with the target's position varied at token scale.** "Lost in the middle" [5] is a large-*token* phenomenon; our original no-decay result was at $N = 20$ record *units*, with the target item at the start (primacy). The token-scale probe (Section 6.1) re-ran it at ∼6.6k tokens with the target (the "needle") at 8% / 50% / 92% depth: no positional decay (30/30, both models; the middle as clean as the ends). We bound this to the tested scale and a *semantically-unique* target; arbitrary token scale and a *camouflaged* target (the regime where [5] bites hardest) remain the standing residual.

- n is small; the scorer is the experimenter (bias runs toward the system, so null results are conservative). Keeping the citations honest was itself a live test of the study's own thesis: a verification pass over the project record found one anchor (a review paper) cited for a claim slightly stronger than a review can support—downgraded to background here, with the load-bearing anchors kept [1,2]—and an *initial over-correction* that labelled it "fabricated" before a re-check showed the identifier was real and correct. Both the weak anchor and the over-correction are instances of the anchor-validity discipline the study formalizes, applied to itself.

For the measurable rows, closure is an experiment; for the unmeasurable rows, closure is this ledger—naming each one precisely, with the reason it resists single-interaction measurement. The study marks the open ground rather than claiming it.

6.1 Experiments — all four run

The literature review pointed to four measurements that would each sharpen a result above. All four were subsequently run (each pre-registered with a locked prediction before any output was read); the open questions are now bounded, not blank.

1. **Three-arm filler control** — *done, five vehicles* (Section 4.6). Re-running with {empty padding, coherent-irrelevant, substance} confirmed the recall (record) KEEP cleanly (10/10 vs 0), moderated the convention KEEP (partly derivable and primable by disposition), and—extended to the **verify** "form" verdict on two vehicles—found it the *most* derivable verdict. A high-stakes completion-gate vehicle hit the ceiling at 40/40 avoid-trap across all four arms (bare included); a *low-stakes* vehicle, run to test whether that ceiling was just the domain priming caution, still showed bare withholding 10/10—caution is derived even away from the high-caution domain, so "legibility, not behaviour" holds. (An apparent residue—a distractor lowering Sonnet's caution to 3/5—did *not* replicate: a rerun at $n = 10$ returned 10/10, Haiku 5/5, so it was noise at $n = 5$; away from that domain the result is purely legibility.) The remaining "form" verdict, **handoff**, was then closed the same way: a structured resume cursor *tied* an equal-length prose summary carrying the same state (10/10 = 10/10), no-state arms 0/30—the structure is legibility, the state content (which prose carries equally) is the substance. No "form" verdict survives the control as a change in behaviour.

2. **Direct empty-filler check on the study's own models** — *done*: empty padding ≈ bare on all probes (pure length did not move these decisions), so filler did not inflate the results here. The conservative-control caveat (§3.2) therefore biases toward *under*-stating the doctrine, as expected.

3. **Cross-scale calibration** — *done* (three-tier ladder, Haiku→Sonnet→Opus, 45 no-fact cells), **with a reproducibility caveat**. Two parts. *Robust:* recall of the unguessable fact does *not*

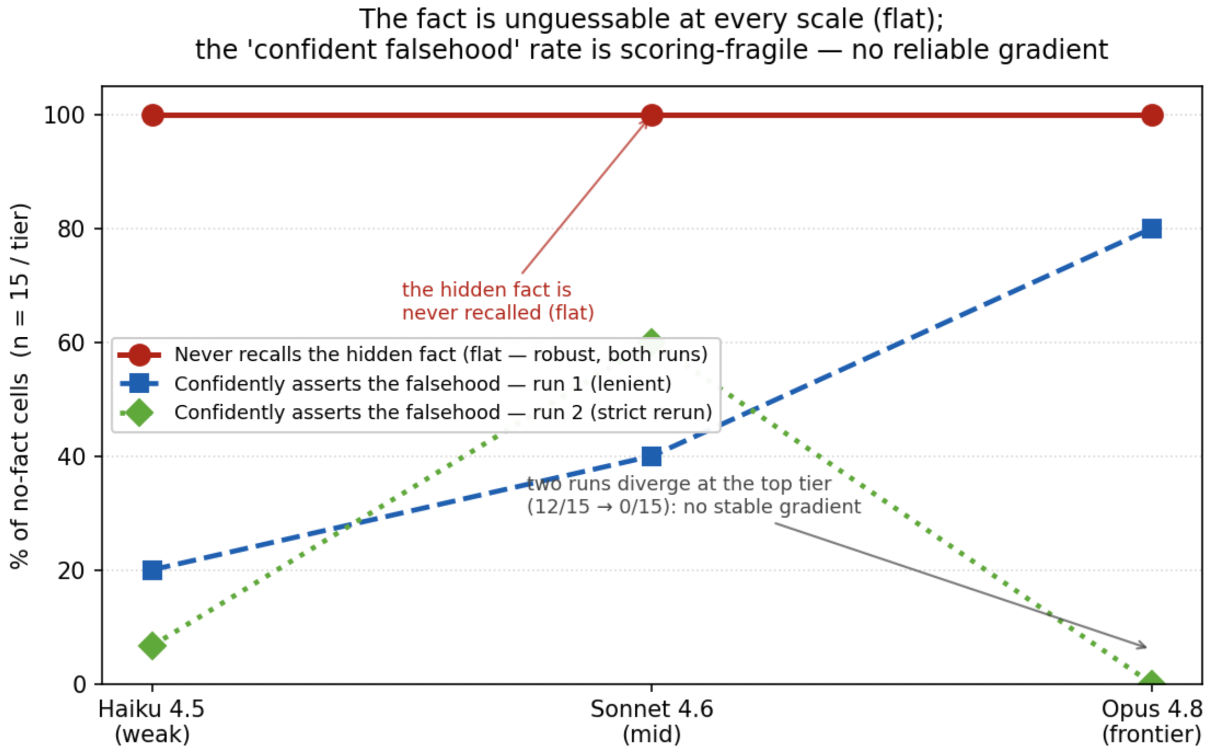

Figure 1: Cross-scale calibration ($n = 15$ / tier), two independent runs. Failing to recall the hidden fact is flat at the ceiling—the fact is unguessable at every capability (robust). The "confidently asserts the falsehood" rate is **fragile to how it is scored**: run 1 (lenient) rises with capability, but run 2 (strict rerun) collapses at the top tier ($12/15 \to 0/15$), because the strongest model pairs the false assertion with a "verify it" step and often rejects the shortcut. There is no reliable capability gradient; the surviving claim is non-elimination, not "stronger $\Rightarrow$ more confident" (SOUL-142).

improve with capability—no tier spontaneously recalls the hidden mechanism (flat across tiers, in both runs), and capability does not *eliminate* the confident false assertion (even the top tier states it in a real fraction of cells). *Not reproducible:* the apparent **gradient**—"confident fabrication rises Haiku < Sonnet < Opus, strongest states it most"—did **not** survive a re-run. The count is fragile to how it is scored: an independent rerun under stricter scoring moved the top tier (Opus) from ~80% to ~0%, because Opus tends to assert the falsehood *and then* recommend verifying it, and rejected the shortcut outright 9/15. So the anchored claim is *non-elimination only*—not a capability gradient (Figure 1; SOUL-141/142).

4. **Token-scale and position depth test** — *done* (the F038 target at 8%/50%/92% depth in a ~6.6k-token record). No positional decay: 30/30 recall at every position, both models; the middle (the worst case per [5]) as clean as the start and end. 'Lost in the middle' did not appear at this scale for a *semantically-unique* target. This bounds the depth claim to the tested scale and a distinctive target; arbitrary scale and a *camouflaged* target (distractors similar to the target, where [5] bites hardest) remain the standing residual.

7 Discussion

The findings come down to one distinction. A meta-doctrine layer can try to make a model *reason better in general*, or it can *carry information the model cannot regenerate.* The first is reproduced, at equal compute, by an equal amount of any content—it is legibility, disposition, or compute, and it fades as the base model improves. The second is not: a fact about a specific API, a convention specific to a project, an incident specific to a team—these have no generic substitute, and they remain decisive at the frontier. And the way the frontier model fails is not benign: a capable model that lacks the fact does not abstain; it can generate a confident, plausible, wrong justification (Section 4.5)—the exact hazard that an external, unguessable anchor is the only defense against. We resist the stronger claim that capability *monotonically* worsens this; our evidence is that capability does not *resolve* it, and that whether the strong or the weak model fails depends on the form of the surviving rule.

This reframes what such systems are *for.* Their durable value is not a reasoning prosthesis but a **memory of the unguessable**, plus the *activation* scaffolding that makes recording and checking reliably happen. The reasoning-ceremony—roles, councils, forced perspectives, structured formats—reads like substance and measures like form. That it reads like substance is not incidental: the same verbosity that makes it *feel* like better thinking is what a length-matched control reproduces.

The compression result sharpens the design rule. Because the dangerous failure is the frontier model fabricating around a missing fact, an always-on rule-set must preserve not just a rule's bare statement but the *force* of the fact behind it—the fact that runs against the model's prior; a terse rule with the incident stripped out invites the very fabrication it was meant to prevent, and a loophole in the rule is the opening the strong model takes.

8 Conclusion

Most of what makes a meta-doctrine system *feel* effective is the system making the work legible and the model more verbose—reproduced, at equal compute, by semantically-empty filler, and fading at the frontier. The part that genuinely changes what the model decides is narrow and holds even for strong models: carrying project-specific, against-the-grain knowledge that the model cannot regenerate and that a generic equivalent gets wrong, together with the scaffolding that makes recording and checking reliably happen. *Derivable dissolves; unguessable persists.* A lean system that keeps the record, a small against-the-grain rule-set, a tested skills catalog, and a completion gate that actually fires—and cuts the reasoning-ceremony to a line—captures the measured value with far less bulk.

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